

Linear model : $y = \beta_0 + \beta_1 X_1 + \epsilon$, $\epsilon \sim N(0, \sigma^2)$

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Generalized Linear Model : $y|x \sim N(\beta_0 + \beta_1 X_1, \sigma^2)$

error
function

X_1 = hours studied Y = receive an A.

X_2 = undergrad GPA $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$

Goal of Generalized Linear Model :

$$Y = -6 + 0.05X_1 + 1 \cdot X_2$$

To make a linear model that can accomoda-

$$a.) X_1 = 40 \quad X_2 = 3.5$$

te continuous, categorical, and non-integer
data.

$$\sigma(\beta_0, \beta_1, \beta_2) = \frac{1}{1+e^{-t}} \quad * \text{Calculating } t$$

$$\sigma(\beta_0, \beta_1, \beta_2) = \frac{1}{1+e^{-(0.5)}} \quad t \rightarrow Y = -6 + 0.05(40) + 1(3.5)$$

$$\sigma(\beta_0, \beta_1, \beta_2) = 0.3775$$

$$t \rightarrow Y = -6 + 2 + 3.5$$

$$t \rightarrow Y = -0.5$$

What random variable that gives the value $0 \leq x \leq 1$?

Bernoulli random variable : $x \sim \text{Bernoulli}(p)$

So, the probability of that student gets an A

$$P(x=0) = 1-P \quad \left\{ \begin{array}{l} P(x=x) = P^x (1-P)^{(1-x)} \\ P(x=1) = P \end{array} \right.$$

in the class is 37.75%.

$$P(x=1) = P$$

$$x = 0, 1$$

Logistic regression : $y|x \sim \text{Bernoulli}(\beta_0 + \beta_1 x)$ b.) $X_1 = ?$ $X_2 = 3.5$ $Y = 0.5$

$$\sigma(\beta_0, \beta_1, \beta_2) = \frac{1}{1+e^{-t}}$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

log-odds = logarithm of odds (probability) ratio

$$0.5 = \frac{1}{1+e^{-t}}$$

$$t=0 \rightarrow 0 = -6 + 0.05(X_1) + 1(3.5)$$

$$0.5(1+e^{-t}) = 1$$

$$0 = -6 + 0.05X_1 + 3.5$$

$$P(X=x) = P^x (1-P)^{1-x} \quad \begin{array}{l} P = \beta_0 + \beta_1 x_i \\ x = y_i \quad (0 \leq y_i \leq 1) \end{array}$$

$$1+e^{-t} = \frac{1}{0.5}$$

$$0 = 0.05X_1 - 2.5$$

$$= (\sigma(\beta_0 + \beta_1 y_i)^{y_i}) (\sigma(\beta_0 + \beta_1 y_i)^{1-y_i})$$

$$e^{-t} = 2 - 1$$

$$2.5 = 0.05X_1$$

$$e^{-t} = 1$$

$$\frac{2.5}{0.05} = X_1$$

$$\log\left(\frac{P}{1-P}\right) = \overbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n}^t$$

$$-t = 0$$

$$50 = X_1$$

$$t = 0$$

$$X_1 = 50 \text{ hours}$$

So, if that student wants to have a 50% chance of
getting an A in the class, he/she has to study for
50 hours

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9.) a.) odds = 0.37 $P(x) = ?$

$$\frac{P(x)}{1-P(x)} = 0.37$$

$$P(x) = 0.37(1-P(x))$$

$$P(x) = 0.37 - 0.37 P(x)$$

$$P(x) + 0.37 P(x) = 0.37$$

$$1.37 P(x) = 0.37$$

$$P(x) = \frac{0.37}{1.37}$$

$$P(x) = \frac{37}{137} \text{ (37 out of 137 people)}$$

$$P + P e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} = e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}$$

$$P e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} - e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} = -P$$

$$e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} (P - 1) = -P$$

$$e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} = \frac{-P}{P-1}$$

$$e^{(\hat{\beta}_0 + \hat{\beta}_1 x)} = \frac{P}{1-P}$$

$$e^{\log\left(\frac{P}{1-P}\right)} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\ln\left(\frac{P}{1-P}\right) = \hat{\beta}_0 + \hat{\beta}_1 x //$$

Since P is probability for orange, then $1-P$ must be the probability for apple

$$\frac{P}{1-P} = \text{odds} \rightarrow \frac{P(Y=\text{orange})}{P(Y=\text{apple})} = \text{odds}$$

b.) $P(x) = 16\%$ odds = ?

$$\text{odds} = \frac{P(x)}{1-P(x)}$$

$$\text{odds} = \frac{0.16}{1-0.16}$$

$$\text{odds} = \frac{0.16}{0.84}$$

$$\text{odds} = \frac{4}{21} = 19.05\%$$

So, the log odds of orange versus apple is %

$$\hat{\beta}_0 + \hat{\beta}_1 x //$$

b.) Log odds of orange versus apple in my friends model = ?

For simplicity, let's assume:

$$e^{(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x)} = m$$

$$e^{(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)} = n$$

$$P(Y=\text{orange} | X=x) = P$$

So, the logistic function of my friend's model will become:

$$P = \frac{m}{m+n}$$

Let's start finding the log odds:

$$P = \frac{m}{m+n}$$

$$m = \frac{-nP}{P-1}$$

$$mP + nP = m$$

$$m = \frac{nP}{1-P}$$

$$mP - m = -nP$$

$$\frac{m}{n} = \frac{P}{1-P}$$

$$m(P-1) = -nP$$

We switch the place

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$$\text{My model} = \hat{P}_r(Y=\text{orange} | X=x) = \frac{e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}{1 + e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}$$

My friend's model:

$$\hat{P}_r(Y | X=x) = \frac{e^{(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x)}}{e^{(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x)} + e^{(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)}}$$

a.) Log odds of orange versus apple in my model = ?

$$\hat{P}_r(Y=\text{orange} | X=x) = \frac{e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}{1 + e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}$$

shorten to P

$$P = \frac{e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}{1 + e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}}$$

$$P(1 + e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}) = e^{(\hat{\beta}_0 + \hat{\beta}_1 x)}$$

$$\frac{P}{1-P} = \frac{m}{n}$$

$$\ln\left(\frac{P}{1-P}\right) = \ln\left(\frac{m}{n}\right)$$

$$\ln\left(\frac{P}{1-P}\right) = \ln(m) - \ln(n)$$

Now let's revert back the m and n:

$$* \ln(m) = \ln(e^{(\hat{\alpha}_{orange0} + \hat{\alpha}_{orange1}u)})$$

$$\ln(m) = \hat{\alpha}_{orange0} + \hat{\alpha}_{orange1}u$$

$$* \ln(n) = \ln(e^{(\hat{\alpha}_{apple0} + \hat{\alpha}_{apple1}u)})$$

$$\ln(n) = \hat{\alpha}_{apple0} + \hat{\alpha}_{apple1}u$$

d.) In my friend's model:

$$\hat{\alpha}_{orange0} = 1.2 \quad \hat{\alpha}_{orange1} = -2$$

$$\hat{\alpha}_{apple0} = 3 \quad \hat{\alpha}_{apple1} = 0.6$$

In my model: $\beta_0 = ?$ $\beta_1 = ?$

$$\beta_0 + \beta_1 u = (\hat{\alpha}_{orange0} - \hat{\alpha}_{apple0}) + (\hat{\alpha}_{orange1} - \hat{\alpha}_{apple1})u$$

$$\beta_0 + \beta_1 u = (1.2 - 3) + (-2 - 0.6)u$$

$$\beta_0 + \beta_1 u = -1.8 - 2.6u$$

$$\text{So, } \beta_0 = -1.8 \text{ and } \beta_1 = -2.6$$

So, the log odds of orange versus apple in

my friend's model is:

$$\ln\left(\frac{P}{1-P}\right) = \hat{\alpha}_{orange0} + \hat{\alpha}_{orange1}u - \hat{\alpha}_{apple0} -$$

$$\hat{\alpha}_{apple1}u$$

$$= (\hat{\alpha}_{orange0} - \hat{\alpha}_{apple0}) + (\hat{\alpha}_{orange1}$$

$$- \hat{\alpha}_{apple1})u$$

e.) Since the log odds of my model and the log

odds of my friend's model in point(d) are equal,

then our predicted class labels will agree 100%

of the time.

$$c.) \text{ in my model } \rightarrow \hat{\beta}_0 = 2, \hat{\beta}_1 = -1$$

in my friend's model $\rightarrow ?$

Since our models are made from the same data,

we can assume that our log odds are equal

$$\beta_0 + \beta_1 u = (\hat{\alpha}_{orange0} - \hat{\alpha}_{apple0}) + (\hat{\alpha}_{orange1} - \hat{\alpha}_{apple1})u$$

To make it more similar, we are going to set the

coefficients for apple to 0, leaving only orange.

$$\beta_0 + \beta_1 u = (\hat{\alpha}_{orange0} - 0) + (\hat{\alpha}_{orange1} - 0)u$$

$$2 + (-1)u = \hat{\alpha}_{orange0} + \hat{\alpha}_{orange1}u$$

$$\text{Thus: } \hat{\alpha}_{orange0} = 2, \hat{\alpha}_{orange1} = -1$$

$$\hat{\alpha}_{apple0} = 0, \hat{\alpha}_{apple1} = 0$$